

Urea Nitrogen

Test Details

METHODOLOGY

Enzymatic (urease)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

High BUN occurs in chronic glomerulonephritis, pyelonephritis and other causes of chronic renal disease; with acute renal failure, decreased renal perfusion (prerenal azotemia) as in shock. With urinary tract obstruction BUN increases (postrenal azotemia), for example as caused by neoplastic infiltration of the ureters, hyperplasia or carcinoma of the prostate. BUN is useful to follow hemodialysis and other therapy. “Uremia” was defined by Luke as an expression of a constellation of signs and symptoms in patients with severe azotemia secondary to acute or chronic renal failure.¹ Causes of increased BUN include severe congestive heart failure, catabolism, tetracyclines with diuretic use, hyperalimentation, ketoacidosis, and dehydration as in diabetes mellitus, but even moderate dehydration can cause BUN to increase. Corticosteroids tend to increase BUN by causing protein catabolism. Bleeding from the gastrointestinal tract is an important cause of high urea nitrogen, commonly accompanied by elevation of BUN:creatinine ratio. Nephrotoxic drugs must be considered.

Borderline high values may occur after recent ingestion of high protein meal and muscle wasting may cause an elevation as well.

With creatinine, BUN is used to monitor patients on dialysis.

Low BUN occurs in normal pregnancy, decreased protein intake, with intravenous fluids, with some antibiotics, and in some but not all instances of liver disease.

As described by DeCaux et al in 1980, in the syndrome of inappropriate secretion of antidiuretic hormone (SIADH): findings include hyponatremia with serum or plasma $\text{Na}^+ \leq 128 \text{ mmol/L}$, hypo-osmolality ($< 260 \text{ mOsm/kg}$ with urine osmolality $> 300 \text{ mOsm/kg}$) with low BUN. Such findings occur in situations in which patients are overhydrated. Clinical findings included absence of edema or evidence of heart, liver, thyroid, renal or adrenal disease.² Hypouricemia, with uric acid levels in 16 of 17 patients $< 4 \text{ mg/dL}$, is reported with the syndrome of inappropriate secretion of antidiuretic hormone.³ (SIADH can be seen with higher serum sodiums and higher osmolalities. Urine osmolality is greater than serum osmolality in SIADH. DeCaux in 1982 presented criteria modified from the 1980 paper.⁴)

Osmolality ($\text{mOsm/kg H}_2\text{O}$) is calculated as follows:

$$\text{Osmolality} = [\text{Na}^+ (\text{mmol/L})] \times [2 + \text{glucose} (\text{mg/dL})] / 18 + [\text{BUN} (\text{mg/dL}) / 2.8]$$

Limitations

Uremia is best evaluated with creatinine as well as urea nitrogen.¹ In both prerenal and postrenal azotemia, for instance, BUN is apt to be increased somewhat more than is creatinine; however, in a series of dehydrated children with gastroenteritis who had metabolic acidosis and increased anion gap, 88% had BUN concentration $\leq 18 \text{ mg/dL}$. The authors found bicarbonate and anion gap more sensitive indices in this setting.⁵ In chronic progressive renal disease, about three-quarters of the renal parenchyma must be damaged or destroyed before azotemia develops.

Custom Additional Information

Although creatinine is generally considered a more specific test to evaluate renal function, they are commonly used together.¹ Luke points out that clinical renal failure is variable between individual patients.¹ Drug effects have been summarized.⁶

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Storage Instructions

Room temperature

Causes for Rejection

Improper labeling

Footnotes

1. Luke RG. Uremia and the BUN. *N Engl J Med*. 1981; 305(20):1213-1215. PubMed 7290133 (<http://www.ncbi.nlm.nih.gov/pubmed/7290133>)
2. Decaux G, Genette F, Mockel J. Hypouremia in the syndrome of inappropriate secretion of antidiuretic hormone. *Ann Intern Med*. 1980 Nov; 93(5):716-717. PubMed 7212483 (<http://www.ncbi.nlm.nih.gov/pubmed/7212483>)
3. Beck LH. Hypouricemia in the syndrome of inappropriate secretion of antidiuretic hormone. *N Engl J Med*. 1979; 301(10):528-530. PubMed 460306 (<http://www.ncbi.nlm.nih.gov/pubmed/460306>)
4. Decaux G, Unger J, Brimiouille S, Mockel J. Hyponatremia in the syndrome of inappropriate secretion of antidiuretic hormone. Rapid correction with urea, sodium chloride, and water restriction therapy. *JAMA*. 1982 Jan 22-29; 247(4):471-474. PubMed 7054549 (<http://www.ncbi.nlm.nih.gov/pubmed/7054549>)
5. Bonadio WA, Hennes HH, Machi J, Madagame E. Efficacy of measuring BUN in assessing children with dehydration due to gastroenteritis. *Ann Emerg Med*. 1989 Jul; 18(7):755-757. PubMed 2735594 (<http://www.ncbi.nlm.nih.gov/pubmed/2735594>)
6. Artur Y, Galimany R. Urea. In: Siest G, Galteau MM, eds. *Drug Effects on Laboratory Test Results Analytical Interferences and Pharmacological Effects*. Littleton, Mass: PSG Publishing Co;1988: 439-453.